

10 of 11 · Roles in the Data Field

Learn

Different roles within the data field contribute to the data analysis process in different ways. Companies often use various titles for these roles, but they function similarly across companies. The increase in the amount of data and technological advancements to store and analyze it has dramatically increased the opportunities across all these roles. The primary roles in the data field are the following:

- **Data Engineer:** data engineers develop, test, and maintain the data architecture. They are involved in the early stages of the data analysis process, making data available to data analysts and data scientists. They typically are part of a company's engineering team. Data engineers have strong technical backgrounds in programming and databases. Adjacent roles include database administrators and data architects.
- **Data Analyst:** data analysts use statistics to organize and analyze data, then extract and communicate insightful information to decision-makers. They typically rely on programming tools like Python, R, and SQL. The Data Analyst title can vary depending on function (e.g., Business Intelligence Analyst, Database Analyst, Marketing Data Analyst, etc.).
- **Data Scientist:** data scientists combine computer science, machine learning, statistics, and math skills to conduct advanced statistical analyses and tests, develop processes for modeling data, and implement algorithms and statistical models to uncover insights from data. Many popular tech products, such as Google Search and Amazon Alexa, are built on algorithms developed by data scientists. They are considered senior to data analysts. Adjacent roles include machine learning engineers and statisticians.
- **Business Analyst:** business analysts translate data insights to help make business decisions. They rely on less technical tools, like Excel, SQL, Power BI, and/or Tableau. They work collaboratively with other data professionals to translate business requirements and data insights. Data analysis is just one of their many functions. More and more non-data roles require the use of data for decision-making, such as financial analysts, product managers, project managers, marketers, and many others. For purposes of this course, we'll classify them as business analysts.

Dataquest has learning paths that help learners build the technical skills to perform well in each of these roles. We encourage you to check them out!